| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 15 | 29.5 or 29.45 to 29.46 | **2** | **M1** for $\frac{60}{360}\times\pi\times7.5^2$ oe |
| 16 | 25 | **2** | **M1** for $x\times\left(1+\frac{6}{100}\right)=26.50$ oe or better |
| 17(a) | 0.1 or $\frac{1}{10}$ | **1** | |
| 17(b) | 90 | **3** | **M2** for $\frac{1}{2}\times10\times2+10\times2+\frac{1}{2}(2+4)\times20$ oe<br>or **M1** for one area calculation or indicated on diagram |
| 18 | 27.15 cao | **3** | **M2** for (9.4 + 0.05) × 2 + 8.2 + 0.05 or better<br>or **M1** for 8.2 + 0.05 or 9.4 + 0.05 or better seen<br>OR<br>**SC2** for answer 25.95<br>or **SC1** for answer 26.85 |
| 19(a) | 61.1 or 61.08 to 61.09... | **3** | **M2** for [$\sin x$ =] $\frac{8\sin100}{9}$ oe or better<br>or **M1** for $\frac{9}{\sin100}=\frac{8}{\sin x}$ oe |
| 19(b) | 11.7 or 11.66 to 11.67 | **3** | **M2** for $\frac{1}{2}\times9\times8\times\sin(180-100-$ *their* **(a)**$)$ oe<br>or **M1** for 180 – 100 – *their* **(a)** |
| 20 | 60 | **3** | **M2** for $4\times\sqrt[3]{\frac{40500}{12}}$ oe<br>or **M1** for $\left(\frac{4}{l}\right)^3=\frac{12}{40500}$ oe<br>or $\sqrt[3]{\frac{40500}{12}}$ oe or $\sqrt[3]{\frac{12}{40500}}$ oe |
| 21(a) | 4 – 2$x$ | **2** | **B1** for 4 or – 2$x$ |
| 21(b) | (2, 10) | **2** | **B1** for $x$-coordinate of 2<br>or **M1** for *their* 4 – 2$x$ = 0 |
| 22(a) | –**a** + **b** | **1** | |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 22(b) | $2\mathbf{a} - \frac{1}{2}\mathbf{b}$ | **3** | **B2** for answer $2\mathbf{a} + p\mathbf{b}$ or $q\mathbf{a} - \frac{1}{2}\mathbf{b}$ $q \neq \frac{1}{2}$ or correct unsimplified answer in terms of **a** and **b**<br><br>or **M1** for $\overrightarrow{AC} = \frac{3}{2}\mathbf{a}$ or $\overrightarrow{OC} = \frac{5}{2}\mathbf{a}$ or correct route<br>If 0 scored **SC1** for answer $\mathbf{a} + \frac{1}{2}\mathbf{b}$ |
| 23 | $\frac{3}{x+1}$ final answer | **3** | **B1** for $2(x+1)-(2x-1)$ oe<br><br>**B1** for common denominator $x + 1$ |
| 24 | (2.4, 1.8) oe | **5** | **M1** for [gradient =] $-1 \div \frac{1}{3}$ oe<br>**M1** for substituting (2, 3) into $y = (their\ m)x + c$ oe<br><br>**M1** for $\frac{1}{3}x + 1 = their\left(mx + c\right)$ with $their\ m \neq \frac{1}{3}$<br><br>**M1** for substituting *their* $x$-coord into either equation to find $y$<br>or for substituting *their* $y$-coord into either equation to find $x$ |
| 25 | 63.4 or 63.43...<br>243.4 or 243.4... | **2** | **B1** for each<br><br>If 0 scored **SC1** for two answers with a difference of 180 |
| 26 | $\frac{x-2}{u+1}$ oe final answer | **4** | **B2** for $(x-2)(u-1)$<br><br>or **B1** for $u(x-2)-(x-2)$ or $x(u-1)-2(u-1)$<br><br>**B1** for $(u-1)(u+1)$ |